

Dinshaw's Kiosk Application: Comprehensive QA & Testing Guide

This guide is designed for QA testers to verify the features, flows, and data transitions of the Dinshaw's Self-Service Kiosk system.

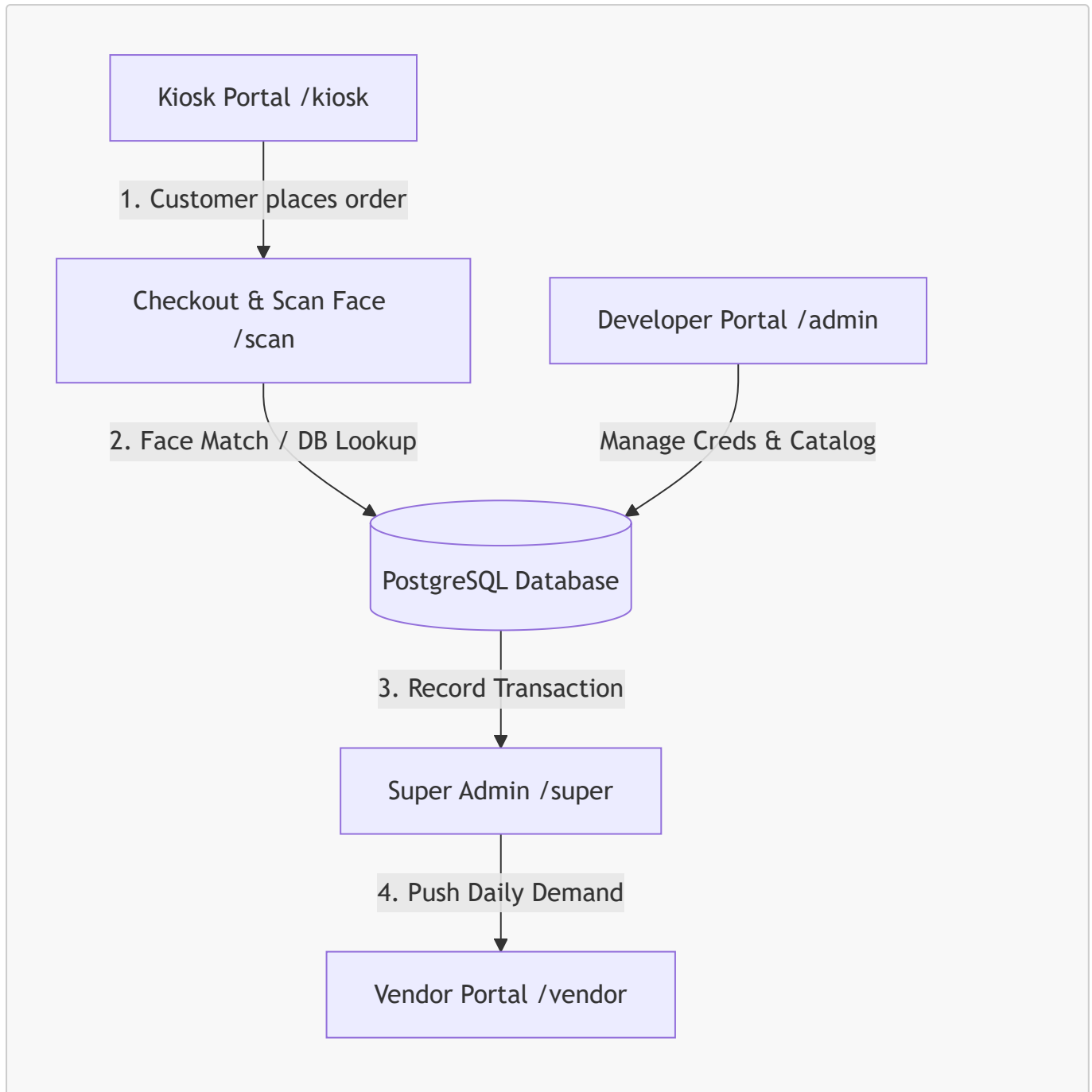
- **Live Deployment Link:** <https://dinshaws.onrender.com>

1. Portal Roles & Default Credentials

The application features a **unified login portal** at `/login`. Logging in with a specific role's credentials automatically routes the user to their respective dashboard.

Portal Role	Target Route	Default Username	Default Password	Purpose
Catalog Developer	<code>/admin</code>	<code>admin</code>	<code>dinshaws</code>	Modifies products, categories, sections, and updates database credentials.
Kiosk Super Admin	<code>/super</code>	<code>superadmin</code>	<code>super123</code>	Views transactions, registers employee faces, and pushes supply demands.
Vendor Portal	<code>/vendor</code>	<code>vendor</code>	<code>vendor123</code>	Accesses pushed daily requirements and downloads CSV orders.

Core Application Architecture & Data Flow



3. 🧪 Step-by-Step Test Suites

Test Suite A: Self-Service Kiosk & Session Timeout (/kiosk)

Goal: Verify product ordering and session inactivity safeguards.

1. **Access the Kiosk:** Navigate to / or /kiosk. You should see the **Touch to Start** welcome screen.
2. **Browse Products:** Click to enter. Browse products using category tabs (Milk, Dahi, Paneer).
3. **Add to Cart:** Click **Add to Cart** on multiple items. Verify the cart badge count increments.
4. **Session Timeout Test (Inactivity):**
 - Leave the screen untouched on the catalog page.
 - **Expected Behavior:** A countdown timer (starts at 15 seconds) will appear in the top-right corner.
 - **Verification:** Wiggle the mouse or tap the screen during the countdown. The timer should disappear and reset.

- **Verification:** Let the countdown reach 0. The cart must be cleared, and you should be redirected to `/timeout`.

5. Session Timeout Immunity:

- Open the **Cart Page** (slide-out cart or full cart page).
 - **Expected Behavior:** No countdown timer should ever appear. The session will remain active indefinitely on the cart page.
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Test Suite B: Face Enrollment & Scan Checkout (`/scan`)

Goal: Verify new user registration and checkout validation.

1. Register a New User:

- Go to Kiosk → Add items to Cart → Click **Checkout**.
- You will be redirected to `/scan`. The camera feed will initialize.
- If your face is not recognized, click **Register/Enroll Face**.
- Enter a test name (e.g., `Tester John`) and a dummy mobile number. Follow the on-screen prompts to capture your face.
- **Expected Behavior:** A success toast appears confirming your enrollment is saved in the database.

2. Scan to Checkout:

- Return to `/scan` with items in the cart.
 - Look directly into the camera.
 - **Expected Behavior:** The system matches your live face against the enrolled database record, displays your name, deducts the order, and redirects to the **Thank You** page.
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Test Suite C: Catalog Developer Portal (`/admin`)

Goal: Verify product catalog management and credential updates.

1. Login: Go to `/login`, enter `admin` / `dinshaws`, and click **Authenticate**.

2. Manage Catalog:

- Add a new product with dummy details.
- Edit an existing product price or image.
- Delete a product.
- **Expected Behavior:** Changes appear instantly on the `/kiosk` page.

3. Change System Credentials:

- Click the **Manage Credentials** button in the header.
 - Modify the Vendor password to `vendor999`.
 - Click **Save**.
 - **Expected Behavior:** A success alert appears. Try logging into `/login` with the new vendor password. It should work instantly.
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Test Suite D: Kiosk Super Admin Portal (`/super`)

Goal: Verify audit logs, face profile administration, and supply push.

1. **Login:** Go to `/login`, enter `superadmin / super123`, and click **Authenticate**.
 2. **Audit Transaction Logs:**
 - Go to the **Transactions** tab.
 - **Expected Behavior:** You should see a list of all transactions completed via the face scan checkout.
 3. **Manage Employees:**
 - Go to the **Employees** tab.
 - **Expected Behavior:** View, edit, or delete enrolled user profiles (like `Tester John` created in Suite B).
 4. **Push Daily Demands:**
 - Go to the **Supply/Vendor** tab.
 - Click **Generate Today's Demand**.
 - Click **Push to Vendor**.
 - **Expected Behavior:** The system aggregates today's transactions, creates a requirement record, and sends it to the Vendor Portal.
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Test Suite E: Vendor Portal (`/vendor`)

Goal: Verify supply chain logs and exports.

1. **Login:** Go to `/login`, enter `vendor / vendor123`, and click **Authenticate**.
 2. **Supply Logs:**
 - **Expected Behavior:** You should see today's pushed requirement cards containing aggregated totals (e.g., `Aahar Milk: 15 units`).
 3. **Export Audits:**
 - Click **Export CSV** on any requirement card.
 - **Expected Behavior:** A `.csv` file downloads containing index, product name, and required quantities.
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Common Troubleshooting on Render Free Tier

- **504 Gateway Timeout or Slow Initial Load:** The server goes to sleep after 15 minutes of inactivity. Please wait 50–90 seconds on the first load for the server to wake up.
- **Camera Access Denied:** Ensure you grant camera permissions in your web browser when visiting the `/scan` checkout or enrollment pages.